

Q. 9
Vector fields $\vec{V}$ & $\vec{W}$ are defined by

$$\vec{V} = (2x - 3y + z, -3x - y + 4z, 4y + z)$$

$$\vec{W} = (2x - 4y - 5z, -4x + 2y, -5x + 6z)$$

check which V.F is Conservative & which is not? Determine if is Conservative & denote it by $\vec{F}$. Find potential fcn ϕ for

Sol

$$\text{Curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-3y+z & -3x-y+4z & 4y+z \end{vmatrix}$$

$$= (0, 1, 0) \neq 0$$

Since $\text{Curl } \vec{V} \neq 0$ so $\vec{F}$ is not Conservative

$$\text{Curl } \vec{W} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-4y-5z & -4x+2y & -5x+6z \end{vmatrix}$$

$$= (0, 0, 0) = 0$$

$\text{Curl } \vec{W} = 0$ so $\vec{F}$ is Conservative.

Suppose $\text{grad } \phi = \vec{W}$

$$\frac{\partial \phi}{\partial x} = 2x - 4y - 5z$$

$$\frac{\partial \phi}{\partial y} = -4x + 2y$$

$$\frac{\partial \phi}{\partial z} = -5x + 6z$$

$$\phi(x, y, z) = \int (2x - 4y - 5z) dx$$

$$\phi(x, y, z) = x^2 - 4xy - 5xz + g(y, z)$$

$$\phi(x, y, z) = \int (-4x + 2y) dy$$

$$\phi(x, y, z) = -4xy + y^2 + h(x, z)$$

$$\phi(x, y, z) = \int (-5x + 6z) dz$$

$$\phi(x, y, z) = -5xz + 3z^2 + k(x, y)$$

$$\phi(x, y, z) = x^2 - 4xy - 5xz + y^2 + 3z^2 + f$$

Q2:- Show that the divergence of inverse-square field

$$\vec{F}(x, y, z) = \frac{c}{(x^2 + y^2 + z^2)^{3/2}} (x\hat{i} + y\hat{j} + z\hat{k})$$

is zero. (Hint let $r = (x^2 + y^2 + z^2)^{1/2}$).

Sol

$$\vec{F}(x, y, z) = \frac{cx\hat{i} + cy\hat{j} + cz\hat{k}}{r^3} = \frac{cx}{r^3}\hat{i} + \frac{cy}{r^3}\hat{j} + \frac{cz}{r^3}\hat{k}$$

$$\text{div } \vec{F} = c \left[\frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) \right]$$

$$\frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) = \frac{r^3 - x(3r^2)(x/r)}{(r^3)^2}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} \quad \frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

$$= \frac{1}{r^3} - \frac{3x^2}{r^5}$$

$$\frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) = \frac{1}{r^3} - \frac{3y^2}{r^5}$$

$$\frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) = \frac{1}{r^3} - \frac{3z^2}{r^5}$$

$$\text{div } \vec{F} = c \left[\frac{3}{r^3} - \frac{3x^2 + 3y^2 + 3z^2}{r^5} \right] = c \left[\frac{3}{r^3} - \frac{3r^2}{r^5} \right] = 0$$

Is it possible for $\vec{F}(x, y) = \langle x^2y, y - xy^2 \rangle$ to be a magnetic field?

5

Q If $\vec{F}$ were magnetic, then its divergence would be zero.

$$= \frac{\partial}{\partial x} (x^2y) + \frac{\partial}{\partial y} (y - xy^2)$$

$$= 2xy + 1 - 2xy$$

$$= 1 \neq 0$$

So $\vec{F}$ cannot model a magnetic field.

Q4:- Is field $\vec{F}(x, y) = \langle x^2y, 5 - xy^2 \rangle$ source free?

Sol The divergence of source free vector field is zero.

$$= \frac{\partial}{\partial x} (x^2y) + \frac{\partial}{\partial y} (5 - xy^2)$$

$$= 2xy - 2xy$$

$$= 0$$

So $\vec{F}$ is source free vector field.

Q5:- Suppose $\vec{v}(x, y) = \langle -xy, y \rangle$, $y > 0$ models the flow of a fluid. Is more fluid flowing into point $(1, 4)$ than flowing out?

Sol We will calculate divergence of $\vec{v}$ at $(1, 4)$

$$\text{div}(\vec{v}) = \frac{\partial}{\partial x} (-xy) + \frac{\partial}{\partial y} (y)$$

$$= -y + 1$$

At $(1, 4)$

$$\text{div}(\vec{v}) = -4 + 1 = -3$$

Since divergence of $\vec{v}$ at $(1, 4)$ is -ive, more fluid is

flowing in than flowing out.

Q6:- Show that gravitational field

$$\vec{F}(x, y, z) = -Gm_1 m_2 \left\langle \frac{x}{(x^2 + y^2 + z^2)^{3/2}}, \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right\rangle$$

has no spin.

Sol

$$P(x, y, z) = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$Q(x, y, z) = \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$R(x, y, z) = \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\begin{aligned} \text{Curl } \vec{F} &= -Gm_1 m_2 [(R_y - Q_z) \hat{i} + (P_z - R_x) \hat{j} + (Q_x - P_y) \hat{k}] \\ &= -Gm_1 m_2 \left[\left(\frac{-3yz}{(x^2 + y^2 + z^2)^{5/2}} - \left(\frac{-3yz}{(x^2 + y^2 + z^2)^{5/2}} \right) \right) \hat{i} \right. \\ &\quad \left. + \left(\frac{-3xz}{(x^2 + y^2 + z^2)^{5/2}} - \left(\frac{-3xz}{(x^2 + y^2 + z^2)^{5/2}} \right) \right) \hat{j} + \left(\frac{-3xy}{(x^2 + y^2 + z^2)^{5/2}} - \left(\frac{-3xy}{(x^2 + y^2 + z^2)^{5/2}} \right) \right) \hat{k} \right] \\ &= 0 \end{aligned}$$

As Curl of Gravitational field is zero so field has no spin.